EXNO-2
5/2/26

DETERMINATION OF CURRENT IN CIRCUIT USING MESH ANALYSIS.

AIM:

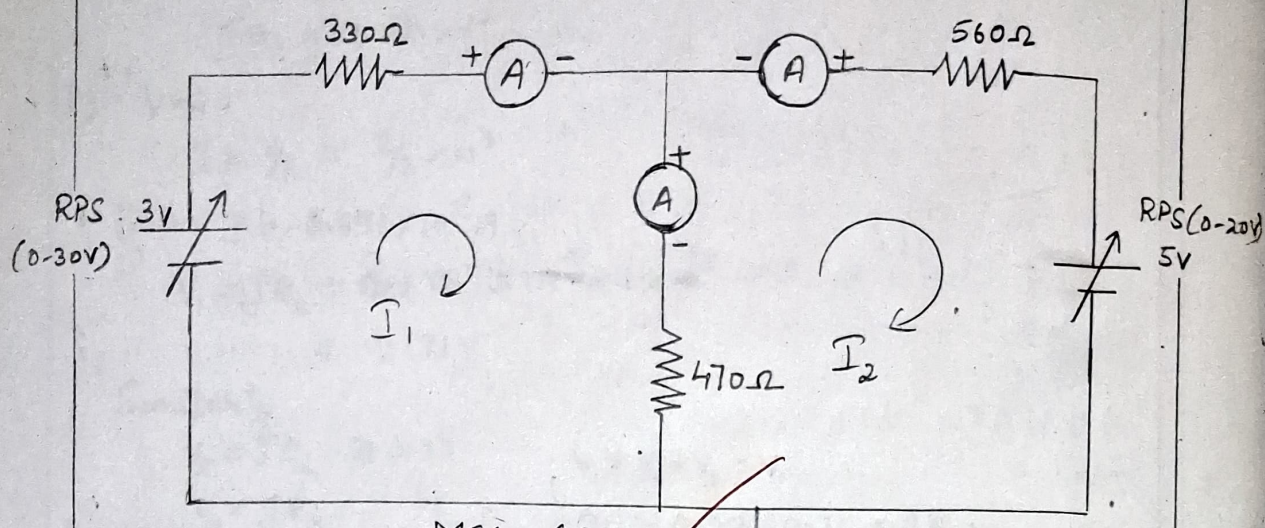
To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED:

S.NO	APPARATUS.	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30 V)	1
2.	Ammeter	(0-30V) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread board.	–	1

PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.
4. Vary the RPS for different input voltages and note down the readings of all the meters.
5. Reduce the RPS to its minimum value and switch OFF the supply.
6. Using the tabulated values, verify Kirchoff's laws practically, verify it theoretically.



MESH CIRCUIT

Parameters	Theoretical	Practical
330Ω	1.2mA	1.3mA
560Ω	4.2mA	4.3mA
470Ω	5.4mA	5.4mA

$I_1 = 2.67$

CALCULATIONS:

$$-3 + 330I_1 + 470(I_1 - I_2) = 0$$

$$330I_1 + 470I_1 - 470I_2 = 3$$

$$800I_1 - 470I_2 = 3 \rightarrow (1)$$

$$560I_2 + 5 + 470(I_2 - I_1) = 0$$

$$560I_2 + 470I_2 - 470I_1 = -5$$

$$-470I_1 + 1030I_2 = 5 \quad -(2)$$

By Cramer's rule for (1) and (2)

$$\begin{bmatrix} 800 & -470 \\ -470 & 1030 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\Delta = 800(1030) - (470)(470)$$

$$= 824000 - 220400$$

$$\boxed{\Delta = 603100}$$

$$\Delta_1 = \begin{bmatrix} 3 & -470 \\ -5 & 1030 \end{bmatrix} = 3090 - 2350$$

$$\boxed{\Delta_1 = 740}$$

$$\Delta_2 = \begin{bmatrix} 800 & 3 \\ -470 & -5 \end{bmatrix} = -4000 + 1410$$

$$\boxed{\Delta_2 = -2590}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{740}{603100} = 0.0012A$$

$$\boxed{I_1 = 1.2mA}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-2590}{603100} = -0.0042A$$

$$\boxed{I_2 = -4.2mA}$$

$$I_3 = I_1 + I_2$$

$$= 1.4 + 4.2$$

$$\boxed{I_3 = 5.4mA}$$

(-ve sign indicates opposite direction)

S.NO	Description	Marks Alotted	Marks Obtained
1.	Experimental Setup / Calculations	5 Marks	20
2.	Observation / Readings.	5 Marks	20
3.	Calculations / Results	5 Marks	20
4.	Viva	5 Marks	20
5.	Record	5 Marks	20
			100

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RESULT:

Thus, the Mesh analysis is verified practically and theoretically. The resultant currents for supply are:

a) The current across 330 Ω resistor is 1.2mA.

b) The current across 560 Ω resistor is 4.2mA.

c) The current across 470 Ω resistor is 5.4mA.